SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code: CSA17	**Course Title:** Artificial Intelligence
CO Assessed: CO1 – Apply (BL3)	**Assessment:** Assessment Tool 1 – Analytical Problem Solving
Weightage: 35% of CIA	**Duration / Marks:** 60 mins
CO1 Statement: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)	
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Section / Batch: _____	**Date:** 3/10/2026

Code of Conduct

I Afrose A , 1924260309 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Afrose A*

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Final answers are presented with units (marks, minutes, percentage points, expected utility); tables and working are shown neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, and rubrics and evaluation page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	Q3
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	Q1, Q2
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	Q1, Q4

Annexure A – Analytical Problem Solving: Answers

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

Given: Squares A and B; start in A, both Dirty; dirt never reappears. Actions: Suck, Right, Left, NoOp. Performance: +1 per clean square at the end of each step, minus p per Left/Right move, over T = 6 steps (p = 1 unless stated). Agent X is a simple reflex agent; Agent Y is a reflex agent with internal state.

(a) PEAS description

Element	Description
Performance measure	+1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move, over a lifetime of T = 6 steps.
Environment	Two squares A and B, each Clean or Dirty. Agent starts in A, both squares Dirty. Dirt never reappears once cleaned.
Actuators	Suck, Left, Right, NoOp.
Sensors	Location sensor (A or B) and dirt sensor (Clean or Dirty) for the current square only.

(b) Trace of Agent X, p = 1

Rule: if the current square is Dirty, Suck; otherwise move to the other square (no memory).

Step	Percept	Action	Clean squares at end	Move penalty	Step score
1	A, Dirty	Suck	1 (A)	0	1
2	A, Clean	Right	1	1	0
3	B, Dirty	Suck	2 (A, B)	0	2
4	B, Clean	Left	2	1	1
5	A, Clean	Right	2	1	1
6	B, Clean	Left	2	1	1
Total			**10**	**4**	**6**

Total score of X (p = 1) = 10 − 4 = 6

(c) Trace of Agent Y, p = 1

Rule: if Dirty, Suck; else if it remembers both squares are clean, NoOp; else move.

Step	Percept	Action	Clean squares at end	Move penalty	Step score
1	A, Dirty	Suck	1 (A)	0	1
2	A, Clean	Right	1	1	0
3	B, Dirty	Suck	2 (A, B)	0	2
4	B, Clean	NoOp (memory: both clean)	2	0	2
5	B, Clean	NoOp	2	0	2
6	B, Clean	NoOp	2	0	2
Total			**10**	**1**	**9**

Total score of Y (p = 1) = 10 − 1 = 9

(d) Scores as functions of p

Both agents collect the same clean-square points: 1 + 1 + 2 + 2 + 2 + 2 = 10. X makes 4 moves and Y makes 1 move.

Score(X) = 10 − 4p Score(Y) = 10 − p

Y is strictly better when 10 − p > 10 − 4p, i.e. 3p > 0, i.e. **p > 0**. At p = 0 both score 10 (tie).

(e) Rationality and interpretation

Rationality depends on the performance measure, the agent's prior knowledge, the actions available, and the percept sequence to date. A rational agent selects the action that maximizes expected performance; it need not be perfect. Since moves cost $p > 0$, wandering after all dirt is gone lowers the score, so **Agent Y is rational for every $p > 0$**. X is rational only in the degenerate case $p = 0$, where wasted moves cost nothing.

At step 5, X sees "A, Clean" and moves, exactly as at step 2. A simple reflex agent maps the current percept alone to an action, so it cannot tell "B still needs cleaning" from "everything is clean". Y's internal state stores the percept history (both squares cleaned), so it stops moving. Both see the same current percepts, but Y uses the percept history and X does not, which is why X cannot match Y.

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

(a) Classification of E1, E2, E3

Property	E1: Autonomous taxi	E2: Part-picking robot	E3: Chess agent (clock)
Observability	Partially observable: blind spots, hidden intentions of other drivers	Partially observable: camera sees only part of the belt, occlusion	Fully observable: whole board visible
Number of agents	Multi-agent: other drivers, pedestrians	Single agent: robot alone on its task	Multi-agent: opponent
Determinism	Stochastic: traffic and road events unpredictable	Stochastic: slips, sensor noise, part variation	Deterministic: a move has a fixed result
Episodic / Sequential	Sequential: current actions affect later states	Episodic: each part judged independently	Sequential: each move shapes the game
Static / Dynamic	Dynamic: world changes while the agent decides	Dynamic: belt keeps moving	Semi-dynamic: board is fixed but the clock runs
Discrete / Continuous	Continuous: position, speed, steering	Continuous: positions and motion	Discrete: finite set of moves and states

(b) Difficulty scores and ranking

Property	E1	E2	E3
Observability	1	1	0
Number of agents	1	0	1
Determinism	1	1	0
Episodic / Sequential	1	0	1
Static / Dynamic	1	1	0.5
Discrete / Continuous	1	1	0
Difficulty score	**6**	**4**	**2.5**

Ranking (hardest to easiest): E1 (6) > E2 (4) > E3 (2.5)

(c) Agent architecture for hardest (E1) and easiest (E3)

E1 (hardest): Utility-based agent (with a learning component). The environment is partially observable, so the agent needs an internal model of the world. It is stochastic, multi-agent, sequential and dynamic, and it must trade off conflicting goals (safety, speed, comfort, fuel). A goal-based agent only distinguishes goal met or not met; a utility function lets the taxi weigh expected outcomes. Learning helps it adapt to new roads and driver behaviour.

E3 (easiest): Goal-based agent. The board is fully observable and moves are deterministic, so the agent can search the game tree toward the goal of checkmate, using an evaluation function to cut off depth within the time limit. No belief state over hidden information is required.

(d) Interpretation

The complexity of the agent program is set by the properties of the task environment, not by the domain label. Partial observability forces the agent to maintain internal state; stochasticity forces it to reason with probabilities and expected utility; a sequential environment forces lookahead and planning; a dynamic environment forces real-time decisions; a multi-agent setting forces it to model other agents; and continuous state and action spaces rule out simple table lookup. A fully observable, deterministic, episodic, static, single-agent environment can be handled by a simple reflex agent, however "intelligent" the domain sounds. Note that chess is a hard game domain but a structurally easy environment, whereas driving is hard on every property.

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes

(a) Correct-identification rate and fooling rate

Correct-identification rate = correct / 50 × 100. Fooling rate = 100% − correct-identification rate (the machine was taken for a human).

System	Correct / Total	Correct-identification rate	Wrong (took machine for human)	Fooling rate
ELIZA (1960s)	47 / 50	94%	3	6%
Chatbot B (2010s)	36 / 50	72%	14	28%
System C (2020s)	29 / 50	58%	21	42%

(b) Turing's 70% criterion and distance from chance

Method: A system meets Turing's criterion if the judge's correct-identification rate is at most 70% (equivalently, fooling rate of at least 30%). Distance from chance = correct-identification rate − 50%.

System	Correct-ID rate	≤ 70%?	Distance from 50% chance
ELIZA	94%	No (fails)	+44 points
Chatbot B	72%	No (fails narrowly)	+22 points
System C	58%	Yes (passes)	+8 points

Only System C satisfies the criterion. As a rough check, for 29/50 against p = 0.5: z = (29 − 25) / √(50 × 0.5 × 0.5) = 4 / 3.54 ≈ 1.13, so System C is not statistically distinguishable from pure guessing at this sample size.

(c) Improvement in fooling rate

ELIZA → System C: 42% − 6% = **36 percentage points**; ratio = 42 / 6 = **7 times**.
Chatbot B → System C: 42% − 28% = **14 percentage points**.

(d) Interpretation

The Turing Test operationalizes only the **acting humanly** approach: it checks whether behaviour in a short text conversation is indistinguishable from a human's. A high fooling rate shows good imitation, which can depend on the judges' biases, the five-minute limit and the conversational tricks used. Searle's **Chinese Room argument** shows that a system can manipulate symbols so convincingly that it passes as a speaker without understanding anything, so passing does not establish "thinking humanly". It also does not establish **acting rationally**, which is defined by maximizing a performance measure (expected utility); imitating human behaviour includes imitating human errors and is not the same as achieving the best outcome. System C's 42% fooling rate is therefore evidence of human-like behaviour, not of thought or rationality.

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

Given: U = 100 − T − 20·L, where L = 1 if the taxi arrives after the 55-minute deadline, else 0.

(a) Utility of each route in each outcome

Route	Outcome	Time T (min)	Late? (T > 55)	Utility U
R1: Highway	Normal	30	No	100 − 30 = 70
R1: Highway	Disrupted (accident)	70	**Late** (L = 1)	100 − 70 − 20 = 10
R2: City road	Normal	45	No	100 − 45 = 55
R2: City road	Disrupted (signal failure)	60	**Late** (L = 1)	100 − 60 − 20 = 20
R3: Ring road	Normal (never disrupted)	50	No	100 − 50 = 50

(b) Expected utilities and rational choice

EU(R1) = 0.70 × 70 + 0.30 × 10 = 49 + 3 = **52**
EU(R2) = 0.90 × 55 + 0.10 × 20 = 49.5 + 2 = **51.5**
EU(R3) = 1.00 × 50 = **50**

The rational agent chooses R1 (Highway), EU = 52 (R1 > R2 > R3).

(c) After the radio alert

P(disruption on R1) becomes 0.80, so P(normal) = 0.20.
EU(R1) = 0.20 × 70 + 0.80 × 10 = 14 + 8 = **22**.
EU(R2) = 51.5 and EU(R3) = 50 are unchanged.

The agent now switches to R2 (City road), EU = 51.5.

(d) Rational versus omniscient

If R1 was chosen and the disruption occurred, the actual utility obtained is U = 100 − 70 − 20 = **10** (late). The agent was **not irrational**: with the information it had, R1 had the highest expected utility (52). A bad outcome from a good decision under uncertainty does not make the decision irrational.

A **rational** agent maximizes *expected* performance given its percept sequence and knowledge. An **omniscient** agent knows the actual outcome of its actions in advance and would maximize *actual* performance (here it would have avoided R1 and taken R2 for 55 or R3 for 50). Omniscience is impossible in practice; rationality is what we can require.

(e) Interpretation

Rationality is defined relative to a performance measure (here U, with its lateness penalty) and to the percept sequence received so far. The radio alert is a new percept: it changes the agent's belief about R1 from 0.30 to 0.80 disruption probability, which drops EU(R1) from 52 to 22. The roads did not change, but the agent's information did, so the expected-utility-maximizing action changed from R1 to R2. A rational agent must use all the evidence it has and update its decision when new percepts arrive.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (100%)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Evaluation Page

Q. No. / Criteria	Problem Understanding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculation (4)	Interpretation of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge

Signature: _____

Date: _____

Course Coordinator

Signature: _____

Date: _____

Program Director

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Date: _____